

Problem #1: Solving Lakota Equations (15 marks)

Here are some equations written in Lakota involving only positive natural numbers:

$$\begin{aligned} \text{wanji} \times \text{nunpa} &= \text{nunpa} \\ \text{yamni} + \text{nunpa} &= \text{zaptan} \\ \text{yamni} + \text{topa} &= \text{nunpa} + \text{zaptan} \\ \text{\textit{\text{šakpe}}} + \text{topa} &= \text{nunpa} \times \text{zaptan} \\ \text{nepcunka} - \text{\textit{\text{šakpe}}} &= \text{yamni} \\ \text{wikcemna} \times \text{ake wanji} &= \text{opawing'e sam wikcemna} \\ \text{wikcemna yamni} + \text{wanji} &= \text{ake wanji} + \text{wikcemna nunpa} \\ \text{\textit{\text{šaglohan}}} - \text{topa} &= \text{topa} \end{aligned}$$

Lakota is a Native American language spoken in the United States by approximately 6,000 speakers living mostly in the northern plains states of North Dakota and South Dakota.

One of the interesting features of the Lakota language is that men exclusively speak certain words, and similarly there are words that are only used by women.

Assignment 1:

Complete the following equations in Lakota.

a) $\text{wikcemna} + \text{nunpa} = \underline{\hspace{2cm}}$

b) $\text{ake wanji} + \text{wanji} = \text{nunpa} + \underline{\hspace{2cm}}$

c) $\text{wikcemna yamni} = \text{zaptan} \times \underline{\hspace{2cm}}$

Assignment 2:

Which numbers do the following Lakota expressions represent?

d) ake topa

e) $\text{opawing'e sam yamni}$

Assignment 3:

Express the following numbers in Lakota?

f) 13

g) 20